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Frequent-voting Independence Screening for Data of Different Types or Different Dimensions

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Abstract: Modern datasets often include different types of variables with complex features, making variable selection particularly challenging. For example, a measure of dependence with the response variable may not be directly comparable among predictor variables of different types and different dimensions. To address this challenge, this work proposes a frequent-voting based independent screening method for variable selection, which avoids a direct comparison of the dependence measure among different variables. Asymptotic analyses show that the proposed method selects all of the active variables with probability converging to one. We also demonstrate its great finite sample performance through numerical experiments and the application to an ADHD study.

Key words and phrases: Model-free; Sure screening; Variable selection; Test of independence

1. Introduction

This work is motivated by collaborative projects with Psychiatrists where the goal of the study is to find risk factors that are predictive of mental disorders such as Attention Deficit Hyperactivity Disorder (ADHD), Major depression, and Suicidal behaviors. This kind of study usually has a large pool of candidate risk factors, consisting of genetic, brain imaging, clinical and demographic variables. Therefore, variable selection plays an important role in these studies. The imaging data such as fMRI data and EEG data are time course data observed at many brain regions. Variable selection usually concerns the selection of brain regions, where the time course data at each region is selected or not selected as a whole. For the gene data analysis, researchers may be interested in selecting relevant gene pathways consisting a group of genes, where the variables (gene pathways) under consideration are multivariate variables. Meanwhile, many clinical data and demographic data are also available in the form of a continuous variable or a categorical variable. In addition, the response variable could be multivariate as well. For example, we may follow subjects for a few time points and produce a longitudinal outcome. These complex data structures often leads to complex nonlinear relationships as well. In consideration of these features, we employ a marginal variable selection approach based on the test of independence for its flexibility in accommodating a combination of scalar, multivariate, and time course data.

With the development of asymptotically consistent methods for the test of independence (Székely et al. (2007); Gretton et al. (2007); Heller et al. (2012); Bergsma and Dassios (2014); Pan et al. (2020)), researchers have naturally considered performing variable selection using dependence measures such as the Distance Correlation measure (Li et al. (2012)) and the Ball Correlation measure (Pan et al. (2019)). In their papers, for each predictor variable in the candidate set, one computes a measure of dependence between the variable under consideration and the response variable, and selects the top N predictor variables where the variables are ordered by the magnitude of the dependence measure. These can be viewed as an extension of the Sure Independence Screening method (SIS) proposed by Fan and Lv (2008). The original SIS and its extensions (Fan et al. (2009); Li et al. (2012); Zhao and Li (2012); Barut et al. (2016)) select variables by ordering the marginal correlation coefficients. The new methods based on dependence measures are nice additions to these existing sure screening methods because dependence measures are well defined for multivariate data and beyond, which are particularly suitable in our application.

To apply dependence-measure-based sure screening method, existing works assume that the magnitude of the estimated dependence measure represents the strength of the variable importance, and the variables with larger values of the dependence measure will be selected. We call these *scale-based* methods. However, we encountered a major challenge when trying to apply these scale-based methods to our studies. When the variables are with different dimensions and in different types, it does not always make sense to determine their importance based on the magnitude of the dependence measure. In simulations we find that irrelevant variables with larger dimensions may have a larger estimated dependence measure and shadow the relevant variables with smaller dimensions. In the ADHD data example, the scale-based method tends to rank many braining imaging variables higher than some important demographic variables. We find it hard to systematically adjust for the dimension of variables in the scale-based method because a sensible adjustment will have to depend on the joint distribution of X and Y and the chosen dependence measure. Although asymptotically the scale-based selection still has sure screening property assuming that the variables with true signals will be separated from the null variables using the chosen measure of dependence, the selection of the empirical cutoff point is particularly difficult when there are variables with different types and dimensions.

To address this challenge in data analysis, we propose a *frequent-voting* method that does not directly compare the magnitude of the dependence measure between different types of data. We order variables by their frequency to pass the independence test in subsamples. Incorporating a resampling procedure into variable selection is intuitively appealing and not a new idea. Meinshausen and Bühlmann (2010) provides some formal analysis on this and illustrates cases where this approach can be combined with various kinds of selection methods. They mention that selecting variables based on "a stability measure", the frequency for each variable to be selected in a random resampling of the data, can better separate relevant variables from irrelevant ones, provides proper guidance to decide the amount of regularization (the size of the model), and sometimes achieves a consistent selection with settings where the original methods fail. We found that this frequent-voting idea is particularly useful when combined with a test of independence.

In Section 2, we formally propose the Frequent-voting Independence Screening method and establish its asymptotic sure screening properties as in previous sure screening methods. We are able to prove that all active variables are selected with probability converging to one, and with appropriate assumptions on the re-sampling ratio, we also show that with probability converging to one none of the inactive variables will be selected. In Section 3, we demonstrate the superb finite sample performance of the proposed method using numerical experiments. In Section 4, we apply the proposed method to the ADHD-200 dataset (part of the 1000 human connectome study). The data consists of one-dimensional phenotype variables and time course fMRI data observed at many brain regions. We show that selection based on the scale-based method may result in selecting redundant brain regions and the frequent-voting method provides more parsimonious and sensible selection results. Proofs for the theorems can be found in the appendix.

2. Frequent-voting independence screening method

Consider a response variable Y and p predictor variables $X_1, \ldots, X_p$. Here Y and $X_k, 1 \le k \le p$ can have different dimensions. The scale-based independence screening will utilize a dependence measure $\rho(X_k, Y)$ as a marginal utility to rank the importance of X_k.

Let X and Y be random variables with marginal distributions P_X on $\mathcal{X}$ and P_Y on $\mathcal{Y}$, respectively, and joint distribution P_{XY} on $\mathcal{X} \times \mathcal{Y}$. We say that X and Y are independent if $P_{XY} \neq P_X P_Y$. A good dependence measure ρ is generally believed to satisfy the following consistency condition.

(C1) $\rho = 0$ if and only if X and Y are independent and $\rho > 0$ otherwise.

Meanwhile, there needs to be a computable empirical counterpart $\hat{\rho}_n$ that satisfies the convergence property,

(C2) Given n independent samples, $n\hat{\rho}_n$ converges to a null distribution when X and Y are independent and diverges to ∞ otherwise.

In recent years, there have been active attempts to develop dependence measures that satisfy the above conditions. The distance covariance measure proposed by Székely et al. (2007) is a widely used measure that satisfy (C1) and (C2). This measure is originally defined for $X \in \mathbb{R}^{d_X}$, $Y \in \mathbb{R}^{d_Y}$ as

$$dCov(X,Y) = \int_{\mathbb{R}^{d_X+d_Y}} ||\phi_{X,Y}(t,s) - \phi_X(t)\phi_Y(s)||^2 w(t,s)dtds$$

where $\phi_X(\cdot)$, $\phi_Y(\cdot)$, $\phi_{X,Y}(\cdot)$ are respective characteristic functions and $w(s,t) = (c_{d_X}c_{d_Y}|t|_{d_X}^{1+d_X}|s|_{d_Y}^{1+d_Y})^{-1}$ with $c_d = \pi^{(1+d)/2}\Gamma\{(1+d)/2\}$. Later it is has been put into a more general formula and works for more general metric spaces (Lyons (2013))

To use this measure for scaled-based independence screening, one need to consider the standardized version

$$dCor(X,Y) = \frac{dCov(X,Y)}{\sqrt{dCov(X,X)dCov(Y,Y)}},$$

ranging from 0 to 1. An empirical version $d\hat{C}or$ can be obtained by plugging in $d\hat{C}ov$. Li et al. (2012) then proposed to rank predictors based on $d\hat{C}or$.

For two univariate normal random variables with the Pearson correlation coefficient r, we can show that $dCor$ is strictly increasing in $|r|$ (Székely et al. (2007)). This property implies that the distance correlation based feature screening procedure is equivalent to the marginal Pearson correlation screening for linear and Gaussian cases. Meanwhile, it is well defined for multivariate variables, and could be more effective than the marginal Pearson correlation screening in the presence of nonlinear relationship, which makes it a good measure to use for independence screening. However, the magnitude of $dCor$ does not always indicate the importance of the predictors when predictors have different types and dimensions. We, therefore, propose to combine a frequent-voting method with the test of independence based on the $dCov$ measure, avoiding a direct comparison of the dependence measure between different variables.

Now consider the variable selection problem for $X_k, k = 1, \ldots, p$. Consider the conditional distribution function $F(y \mid X) = F(Y \leq y \mid X)$. We have the following definition of index sets:

$$\mathcal{A} = \{k : F(y \mid X_k) \text{ functionally depends on } X_k \text{ for some } y\},$$

$$\mathcal{I} = \{k : F(y \mid X_k) \text{ does not functionally depends on } X_k \text{ for any } y\}.$$

Predictors in $\mathcal{A}$ are called *active* predictors. Predictors in $\mathcal{I}$ are called *inactive* predictors. We use p^* to denote the cardinality of $\mathcal{A}$. With a sample of size n, our aim is to construct $\hat{\mathcal{A}}_n$ that contains all active predictors and contains none or very few inactive predictors.

For $k = 1, ..., p$, let $\rho_k = dCov(X_k, Y)$, and $\tau_k = \tau(X_k, Y)$ as the α-level critical value from the null-distribution. We use the notation $z_{ki} = (X_{ki}, Y_i)$ to denote the paired data. For any given data $Z = (z_{ki_1}, ..., z_{ki_m})$ with size m, $\hat{\rho}_{k;m}$ is the empirical $dCov$ based on the m pairs of data. For a given sub-sample size m, we consider all ordered subset of m different integers chosen from $\{1, ..., n\}$, and its corresponding subsamples $z_{ki_1}, ..., z_{ki_m}$. A statistical decision based on m subsamples $z_{ki_1}, ..., z_{ki_m}$ is expressed as

$$h(z_{ki_1}, ..., z_{ki_m}; \tau_k) = I(m\hat{\rho}_{k;m} > \tau_k).$$

A decision that takes value 1 is called a vote for the k-th variable. The selection of the k-th variable is based on the frequency vote it gets from the test of independence using multiple subsamples of size m.

Formally, the total frequency vote of X_k from all possible m tuples is defined as

$$V_{n;m}^k = \binom{n}{m}^{-1} \sum_{i_1<...<i_m} h(z_{ki_1}, ..., z_{ki_m}; \tau_k), \tag{2.1}$$

where the summation is taken over all sets of m different integers chosen from $\{1, ..., n\}$. For a pre-specified frequency $\theta \in (\alpha, 1)$, the selected set of variables is then

$$\hat{\mathcal{A}}_n = \{k : V_{n;m}^k > \theta\}. \tag{2.2}$$

In practice, it is desirable to approximate $V_{n;m}^k$ with some large enough number of subsamples rather than computing all n-choose-m combinations.

This method does not compare $\hat{\rho}_k$ between different variables, but only compare the measure $\hat{\rho}_k$ with its own critical value of the null distribution. The cutoff frequency θ may be determined by the user according to the application. Our numerical studies show that the results are relatively stable across different choices of the frequency thresholds, 50, 70, 80, or 90% of total subsamples.

It is possible to use other dependence measures with a standardized version in the scale-based independence screening. Pan et al. (2019) proposed to utilize the ball correlation measure (Pan et al. (2020)) to rank the importance of variables. It presents similar problems when comparing predictors with different dimensions. While the numerical experiments and data analysis focuses on the use of the distance covariance test *dCov* and the distance correlation measure $dCor$, the frequent-voting framework and the following asymptotic analysis does not tie to a particular method of independence test as long as it satisfies the technical conditions.

The following assumptions are needed to prove sure screening properties.

(A1) There exist some constant c_1 and $0 < \kappa < 1/2$ such that

$$\min_{k \in \mathcal{A}} \rho_k \geq c_1 n^{-\kappa}.$$

(A2) For any $\epsilon > 0$, there exists some constant c_2 such that

$$P(|\hat{\rho}_{k,n} - \rho_k| > \epsilon) \leq \exp(-c_2 n \epsilon^2).$$

Assumption (A1) states that the dependency between each active predictor and the response variable should not be too weak. It is similar to (C2) of Li et al. (2012) and (C1) of Pan et al. (2019). Assumption (A2) ensures that the difference between the population dependency and the estimated statistic is bounded by an exponential function. This property has been shown to hold for distance covariance and ball covariance as described in the appendices of Li et al. (2012) and Pan et al. (2019).

Theorem 1. *(Sure screening property) Assume (A1)-(A2). Let* $m = c_0 n^{\gamma}$ *for* $\gamma \in (2\kappa, 1]$*, and a constant* $c_0 > 0$*. If* $log(p^*) = o(n^{(\gamma - 2\kappa) \vee (1-\gamma)})$*, we have* $P(\mathcal{A} \subset \hat{\mathcal{A}}_n) \xrightarrow{p} 1$.

Theorem 1 guarantees that all active predictors are selected in the model with probability converging to one, when p^*, the cardinality of $\mathcal{A}$, grows at a rate slower than certain exponential function of n, which is easy to satisfy.

Theorem 2. *Assume (A1)-(A2). Let* $m = c_0 n^\gamma$ *for* $\gamma \in (2\kappa, 1)$ *and a constant* $c_0 > 0$. *If* $log(p - p^*) = o(n^{1-\gamma})$, *we have* $P(\hat{\mathcal{A}}_n \subset \mathcal{A}) \xrightarrow{p} 1$.

Theorem 2 requires m to be in a smaller order than n. When $m = c_0 n$, we can compute an upper bound of $P(k \in \hat{\mathcal{A}}_n | k \in \mathcal{I})$ using a Chebyshev inequality;

$$P(V_{n;m}^k > \theta) = P(V_{n;m}^k - E[V_{n;m}^k] > \theta - E[V_{n;m}^k]) \leq \frac{Var[V_{n;m}^k]}{2(\theta - E[V_{n;m}^k])^2}.$$

Based on the U-statistic result $Var[V_{n;m}^k] \leq m/n \cdot Var[h^k]$ (Serfling (2009)), the bound asymptotically becomes $c_0\alpha(1-\alpha)/2(\theta-\alpha)^2$. When $\alpha = 0.05$, $c_0 = 0.8$ and $\theta = 0.8$, this is approximately 0.03.

3. Simulation

We consider $p = 200$ covariates. $X_1, ..., X_{100}$ are one-dimensional and generated from a normal distribution with zero mean and a covariance matrix $\Sigma = (\sigma_{ij})_{100\times100}$ where $\sigma_{ij} = 0.5^{|i-j|}$. The rest $X_{101}, ..., X_{200}$ are eleven-dimensional, where variables in each dimension are normally distributed with the same covariance matrix Σ. We conduct simulations for three sample sizes $n = 100, 150, 200$.

Examples 1-3 are for a continuous response Y, and designed to demonstrate different types of relationships. The notation $\overline{X}$ denotes the average of the eleven-dimensional components of X and $\overline{X^2}$ denotes the average of the squared dimensions.

Example 1. $Y = 1.2X_1 + 1.2X_{11} + 8\overline{X_{101}} + 8\overline{X_{111}} + \epsilon$, $\epsilon \sim N(0, 1)$.

Example 2. $Y = 0.8X_1 + X_{11}^2 + 4\overline{X_{101}} + 3I(\overline{X_{111}} > 0) + \epsilon$, $\epsilon \sim N(0, 1)$.

Example 3. $Y = 1.2X_1 + X_{11}^2 + 5\overline{X_{101}} + 5\overline{X_{111}^2} + \epsilon$, $\epsilon \sim N(0, 1)$.

Example 4 is designed to evaluate the performance of the methods when the response variable Y is binary. Y follows a Bernoulli distribution with probability $1/(1 + e^{\pi})$ where π is generated as follows:

Example 4. $\pi = X_1 + X_2 + 5\overline{X_{101}} + 5\overline{X_{111}}$.

The results from 500 simulations are summarized in Tables 1 - 4.

For the scale-based method, normalized versions $dCor(X_k, Y)$ for $k = 1, ..., p$ are used. There is no universal way to determine the model size of the scale-based method and previous papers (Li et al. (2012); Pan et al.

Table 1: The proportions that each or all active predictors are selected in Example 1. Size is the average number of selected predictors with given frequency cutoffs

		n=100				n=150				n=200			
		90%	80%	70%	50%	90%	80%	f0%	50%	90%	80%	70%	50%
Freq	$\mathcal{P}_1$	0.52	0.62	0.70	0.79	0.76	0.83	0.86	0.91	0.88	0.92	0.93	0.97
	$\mathcal{P}_{11}$	0.51	0.61	0.68	0.77	0.71	0.79	0.83	0.89	0.87	0.89	0.93	0.96
	$\mathcal{P}_{101}$	0.85	0.89	0.92	0.95	0.98	0.98	0.99	0.99	1.00	1.00	1.00	1.00
	$\mathcal{P}_{111}$	0.82	0.88	0.91	0.95	0.97	0.98	0.99	0.99	1.00	1.00	1.00	1.00
	$\mathcal{P}_{all}$	0.23	0.33	0.41	0.56	0.51	0.64	0.70	0.80	0.76	0.81	0.86	0.93
Scale	$\mathcal{P}_1$	0.28	0.31	0.34	0.38	0.49	0.52	0.55	0.59	0.68	0.72	0.74	0.78
	$\mathcal{P}_{11}$	0.27	0.30	0.32	0.37	0.46	0.49	0.51	0.56	0.69	0.72	0.74	0.77
	$\mathcal{P}_{101}$	0.92	0.96	0.97	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	$\mathcal{P}_{111}$	0.90	0.95	0.97	0.99	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	$\mathcal{P}_{all}$	0.07	0.09	0.11	0.15	0.24	0.27	0.30	0.35	0.47	0.52	0.55	0.60
Size		4.5	6.4	8.5	13.8	5.1	6.5	7.8	11.3	6.8	8.5	10.3	15.1

(2019)) have suggested to use $(n/logn)$, which are much larger then the true model size in our simulations. For comparison purpose, we list the selection results for multiple model sizes that match the size used in the frequent-voting method. We hope to retain the true variables with high probability with a reasonable size of the model. For the frequent-voting method, the results are based on 500 subsamples with a significance level of 0.05 for the independence $dCov$ test. Subsampling ratio 0.8 is used. The l_2

Table 2: The proportions that each or all active predictors are selected in Example 2. Size is the average number of selected predictors with given frequency cutoffs

		n=100				n=150				n=200			
		90%	80%	70%	50%	90%	80%	70%	50%	90%	80%	70%	50%
Freq	$\mathcal{P}_1$	0.48	0.59	0.64	0.77	0.76	0.83	0.88	0.93	0.90	0.94	0.96	0.97
	$\mathcal{P}_{11}$	0.47	0.55	0.64	0.72	0.81	0.88	0.91	0.95	0.89	0.92	0.95	0.97
	$\mathcal{P}_{101}$	0.52	0.66	0.70	0.76	0.85	0.89	0.93	0.96	0.92	0.96	0.96	0.98
	$\mathcal{P}_{111}$	0.69	0.78	0.84	0.90	0.98	0.98	0.99	1.00	1.00	1.00	1.00	1.00
	$\mathcal{P}_{all}$	0.14	0.24	0.32	0.45	0.51	0.63	0.71	0.84	0.76	0.84	0.88	0.92
Scale	$\mathcal{P}_1$	0.24	0.28	0.31	0.34	0.55	0.58	0.59	0.64	0.68	0.72	0.75	0.78
	$\mathcal{P}_{11}$	0.13	0.14	0.17	0.23	0.20	0.26	0.30	0.38	0.48	0.49	0.53	0.59
	$\mathcal{P}_{101}$	0.63	0.72	0.78	0.84	0.92	0.93	0.96	0.98	0.95	0.98	1.00	1.00
	$\mathcal{P}_{111}$	0.80	0.86	0.89	0.95	0.99	0.99	1.00	1.00	1.00	1.00	1.00	1.00
	$\mathcal{P}_{all}$	0.02	0.04	0.06	0.09	0.08	0.12	0.15	0.24	0.29	0.33	0.38	0.45
Size		4.4	7.2	9.8	17.0	5.8	8.1	10.5	17.0	5.8	7.6	9.5	14.5

distance is used in calculating the distance covariance. Critical values are approximated by a permutation procedure on the whole data set with size n. The total number of variables selected, some of which could be multiple dimensional, is determined by a frequency cutoff; variables are selected if they achieve 90%, 80%, 70% or 50% of the vote from subsampled data. The results are based on the implementation using R packages "dcov".

In each table, the twelve columns represent the results under four dif-

Table 3: The proportions that each or all active predictors are selected in Example 3. Size is the average number of selected predictors with given frequency cutoffs

		n=100				n=150				n=200			
		90%	80%	70%	50%	90%	80%	70%	50%	90%	80%	70%	50%
Freq	$\mathcal{P}_1$	0.70	0.78	0.81	0.89	0.93	0.95	0.96	0.98	0.99	0.99	0.99	1.00
	$\mathcal{P}_{11}$	0.19	0.27	0.33	0.45	0.45	0.55	0.61	0.74	0.74	0.81	0.85	0.90
	$\mathcal{P}_{101}$	0.44	0.55	0.61	0.71	0.75	0.83	0.87	0.92	0.93	0.96	0.97	0.98
	$\mathcal{P}_{111}$	0.16	0.23	0.32	0.44	0.37	0.45	0.54	0.66	0.65	0.76	0.81	0.87
	$\mathcal{P}_{all}$	0.03	0.07	0.10	0.20	0.18	0.26	0.34	0.49	0.49	0.62	0.68	0.78
Scale	$\mathcal{P}_1$	0.51	0.53	0.57	0.61	0.80	0.82	0.83	0.86	0.92	0.92	0.94	0.95
	$\mathcal{P}_{11}$	0.03	0.03	0.04	0.07	0.06	0.07	0.08	0.10	0.16	0.20	0.24	0.30
	$\mathcal{P}_{101}$	0.58	0.68	0.76	0.84	0.85	0.92	0.94	0.97	0.97	0.99	0.99	1.00
	$\mathcal{P}_{111}$	0.22	0.33	0.43	0.57	0.52	0.64	0.74	0.83	0.81	0.89	0.92	0.97
	$\mathcal{P}_{all}$	0.00	0.00	0.01	0.03	0.03	0.04	0.05	0.07	0.12	0.16	0.21	0.27
Size		2.5	4.1	5.7	10.2	3.7	5.1	6.5	10.8	5.0	6.6	8.3	12.9

ferent frequency cutoffs and with three levels of samples sizes. We evaluate the performance through $\mathcal{P}_1$, $\mathcal{P}_{11}$, $\mathcal{P}_{101}$, $\mathcal{P}_{111}$ and $\mathcal{P}_{\text{all}}$, which are the proportions of retaining each active variable and all of the active variables in 500 replications. As expected, we observe that the average size of the model increases as the frequency cutoff becomes lower. The performance of the frequent-voting method and the scale-based method improves as the sample size increases. Overall, the performance of the frequent voting-method is

Table 4: The proportions that each or all active predictors are selected in Example 4. Size is the average number of selected predictors with given frequency cutoffs

		n=100				n=150				n=200			
		90%	80%	70%	50%	90%	80%	70%	50%	90%	80%	70%	50%
Freq	$\mathcal{P}_1$	0.37	0.45	0.52	0.63	0.60	0.68	0.74	0.84	0.76	0.83	0.87	0.92
	$\mathcal{P}_{11}$	0.38	0.46	0.53	0.63	0.60	0.69	0.75	0.84	0.79	0.83	0.87	0.92
	$\mathcal{P}_{101}$	0.31	0.42	0.49	0.62	0.63	0.74	0.79	0.87	0.86	0.90	0.92	0.96
	$\mathcal{P}_{111}$	0.27	0.36	0.43	0.57	0.64	0.72	0.78	0.85	0.86	0.92	0.95	0.97
	$\mathcal{P}_{all}$	0.00	0.01	0.03	0.12	0.12	0.20	0.31	0.49	0.41	0.56	0.65	0.78
Scale	$\mathcal{P}_1$	0.32	0.36	0.40	0.44	0.51	0.56	0.61	0.65	0.68	0.72	0.74	0.78
	$\mathcal{P}_{11}$	0.34	0.38	0.41	0.44	0.53	0.57	0.60	0.63	0.72	0.76	0.78	0.81
	$\mathcal{P}_{101}$	0.37	0.50	0.59	0.72	0.69	0.81	0.86	0.93	0.89	0.93	0.95	0.98
	$\mathcal{P}_{111}$	0.31	0.45	0.55	0.70	0.69	0.80	0.84	0.90	0.89	0.95	0.97	0.99
	$\mathcal{P}_{all}$	0.00	0.01	0.03	0.08	0.11	0.18	0.22	0.31	0.36	0.47	0.51	0.60
Size		2.1	3.1	4.3	7.4	3.6	4.9	6.2	9.5	4.6	5.8	7.2	10.3

very satisfactory with a sufficient sample size $n = 200$, in the sense that the probability of retaining all important variable is close to 1 with a reasonable model size. In all of the settings, the frequent-voting method is always better than the scale-based method, in the sense that the probability of retaining all true variables is higher with the frequent-voting method when the size of the model is kept the same. We can see that the scale-based method tends to miss the one-dimensional active variables X_1 and X_2, and include inactive variables with higher dimensions.

4. Application: ADHD 200

In this section, our method is employed to identify important variables related to Attention Deficit Hyperactivity Disorder (ADHD). ADHD is a common neurological disorder prevalent among school-aged children (Willcutt (2012)), characterized by difficulties in attention and impulse control. We use ADHD-200 consortium dataset (ADHD-200-Consortium (2012); Bellec et al. (2017)), which was publicly released to support the development of scientific tools for diagnosing the condition. As this dataset was made available as part of the ADHD-200 Global Competition, it naturally underwent a rigorous preprocessing step, making it highly suitable for methodological research as evidenced by previous publications.

The dataset contains phenotype variables and resting-state fMRI data written into MNI space at 4 mm x 4 mm x 4 mm voxel resolution. In each voxel, the mean blood-oxygen-level dependent (BOLD) signal was recorded at equally spaced time points. The data is processed using the Athena pipeline and aggregated over functionally parcellated regions of interest (ROIs) called "CC200" (Craddock et al. (2012)). The "ADHD Rating Scale IV" measurement is used as the response variable, which represents the severity of symptoms on a continuous scale. This analysis focuses on data collected at the NYU site, resulting in a final sample of 215 observations with seven phenotype variables and time course fMRI data from 190 ROIs. The phenotype variables include gender, age, handedness, verbal IQ, full-scale IQ, performance IQ, and medication status, all of which are one-dimensional.

To use the distance covariance test *dCov* or the distance correlation measure *dCor* in screening, one needs to employ an appropriate distance measure. For the phenotype data, l_2 distance is used. For the rs-fMRI data, we choose not to apply the l_2 distance directly on the time-series data, because the original time-series data are noisy and contain individual level horizontal shift, which could lead to spurious distance between pairs of ROIs. Following some previous papers (Biswal et al. (1995); Yu-Feng et al. (2007)) in ADHD studies with fMRI data, we transformed each time-series data to *Global Wavelet Power Spectrum* (GWPS) and then applied l_2 distance on the coefficients. The GWPS transformation effectively summarizes the data by indicating the average power at specific frequency levels. The frequency of interest in this data is from 0 to 0.25 Hz, on an equally spaced grid of 60. Morlet's wavelet (R package "biwavelet") is used for our analysis. In addition to the curve analysis, we also consider a five-dimensional summary for the fMRI data, where data in each ROI is summarized into average GWPS values in five predefined frequency bins (0-0.0117 Hz, 0.0117-0.0273 Hz, 0.0273-0.0742 Hz, 0.0742-0.1992 Hz, 0.1992-0.25 Hz) (Zhang et al. (2015); Wang et al. (2015); Luo et al. (2020)). Then l_2 distance is used for this five-dimensional data when applying the $dCov$ or $dCor$ methods.

Both the frequency-voting method and the scale-based method are applied to select variables using distance covariance/distance correlation measures. The implementation details are the same as specified in the simulation section. For the phenotype variables, the frequency vote of each variable from the subsamples is summarized in Table 5. The ranking of variables based on the frequency vote exactly matches the ranking based on the magnitude of $dCor$ (scale-based method). This is expected as both methods provide a proper ordering of variables when the variables have the same dimension. The order of variables is also aligned with the p-values from an individual test of independence between each predictor and the response variable using $dCov$. Several studies reveal an association between "Full-scale IQ" and ADHD (Bridgett and Walker (2006); Fabio et al. (2022)) and the p-value from the test of independence also supports this result. Therefore, the "Full-scale IQ" is used as a cutoff for the rest of

Table 5: Summary of the association between phenotype variables and the response

Variable	Frequency vote	dCor	P-value
Medication Status	100%	0.49	0.00
Gender	100%	0.27	0.00
Verbal IQ	76%	0.20	0.01
Age	74%	0.18	0.02
Full-scale IQ	54%	0.17	0.04
Performance IQ	11%	0.14	0.12
Handedness	4%	0.12	0.17

the analysis. This means that variables, phenotypes and brain ROIs, are selected if their votes exceed 54% for the frequent-voting method and if their dependence measure ($dCor$) exceed that of "Full-scale IQ" for the scale-based method.

The results of the brain ROI selection are summarized in Table 6 and Figure 1. The regions are ordered by the frequency vote or the magnitude of $dCor$. Each number represents a brain region labeled by CC200 parcellation. Both the curve data and its five-dimensional summary data are used. The result from the frequent-voting method remains stable for both versions of the data, where eight out of the nine selected brain regions

Table 6: Selection path of brain ROIs using Full-scale IQ as a cutoff with CC200 labels. Numbers of selected ROIs are inside the parenthesis

Method	Data	ROI selection path
Frequent-voting	5-dim	138 44 33 112 1 81 35 36 31 (9)
	Curve	138 44 33 1 35 189 31 112 81 (9)
Scale-based	5-dim	138 33 44 112 1 35 31 81 102 160 175 90 119 122 36 (15)
	Curve	138 33 35 112 1 31 44 102 160 90 32 189 122 119 81 175 108 173 47 (19)

are the same. The scale-based method selects more brain regions. If the curve data are used, the scale-based method selects an even larger number of brain regions comparing to the result based on the five-dimensional version. We find that the ROIs selected by the frequent-voting method are a subset of ROIs selected by the scale-based method. Our simulations under a similar setting showed that the scale-based method tends to rank high-dimensional inactive variables higher than some low-dimensional active variables. While it is difficult to directly verify the correctness of the selection in this data analysis, we found that the frequent-voting method provides more parsimonious and stable results.

The top ROIs selected by both methods have been identified and well

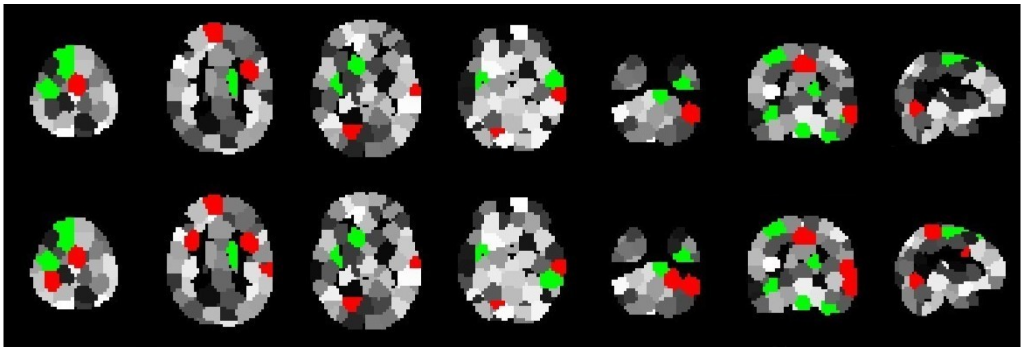

Figure 1: The selected brain regions are highlighted in green if selected by both the scale-based and the frequent-voting methods, and in red if only selected by the scale-based method. The top row displays the results using five-dimensional summary data; while the bottom row shows the results using the curve data. The brain images include five different horizontal slices, one sagittal slice, and one coronal slice.

discussed in ADHD literature. For example, ROI 138 is mainly comprised of *Right Cerebellum* and *Fusiform Gyrus* which has been confirmed as related to ADHD with different data and theories (Wolf et al. (2009); Lei et al. (2014); Stoodley (2016); Chiang et al. (2020)). ROI 33 is largely a part of *left superior temporal gyrus* and *left supramarginal gyrus* which are found in Rubia et al. (2007); Wolf et al. (2009); Sidlauskaite et al. (2015); Zhang et al. (2020). ROI 44 contains a part of *lingual gyrus*, as discussed in An et al. (2013); Zhao et al. (2017); Lan et al. (2021).

References

ADHD-200-Consortium (2012). The adhd-200 consortium: a model to advance the translational potential of neuroimaging in clinical neuroscience. *Frontiers in systems neuroscience 6*, 62.

An, L., Q.-J. Cao, M.-Q. Sui, L. Sun, Q.-H. Zou, Y.-F. Zang, et al. (2013). Local synchronization and amplitude of the fluctuation of spontaneous brain activity in attention-deficit/hyperactivity disorder: a resting-state fmri study. *Neuroscience bulletin 29*(5), 603–613.

Barut, E., J. Fan, and A. Verhasselt (2016). Conditional sure independence screening. *Journal of the American Statistical Association 111*(515), 1266–1277.

Bellec, P., C. Chu, F. Chouinard-Decorte, Y. Benhajali, D. S. Margulies, and R. C. Craddock (2017). The neuro bureau adhd-200 preprocessed repository. *Neuroimage 144*, 275–286.

Bergsma, W. and A. Dassios (2014). A consistent test of independence based on a sign covariance related to kendall's tau. *Bernoulli 20*(2), 1006–1028.

Biswal, B., F. Zerrin Yetkin, V. M. Haughton, and J. S. Hyde (1995).

Functional connectivity in the motor cortex of resting human brain using echo-planar mri. *Magnetic resonance in medicine 34*(4), 537–541.

Bridgett, D. J. and M. E. Walker (2006). Intellectual functioning in adults with adhd: a meta-analytic examination of full scale iq differences between adults with and without adhd. *Psychological assessment 18*(1), 1.

Chiang, C.-T., C.-S. Ouyang, R.-C. Yang, R.-C. Wu, and L.-C. Lin (2020). Increased temporal lobe beta activity in boys with attention-deficit hyperactivity disorder by loreta analysis. *Frontiers in Behavioral Neuroscience 14*, 85.

Craddock, R. C., G. A. James, P. E. Holtzheimer III, X. P. Hu, and H. S. Mayberg (2012). A whole brain fmri atlas generated via spatially constrained spectral clustering. *Human brain mapping 33*(8), 1914–1928.

Fabio, R. A., G. E. Towey, and T. Caprì (2022). Static and dynamic assessment of intelligence in adhd subtypes. *Frontiers in Psychology*, 712.

Fan, J. and J. Lv (2008). Sure independence screening for ultrahigh dimensional feature space. *Journal of the Royal Statistical Society: Series B (Statistical Methodology) 70*(5), 849–911.

Fan, J., R. Samworth, and Y. Wu (2009). Ultrahigh dimensional feature selection: beyond the linear model. *The Journal of Machine Learning Research 10*, 2013–2038.

Gretton, A., K. Fukumizu, C. Teo, L. Song, B. Schölkopf, and A. Smola (2007). A kernel statistical test of independence. *Advances in neural information processing systems 20*.

Heller, R., Y. Heller, and M. Gorfine (2012). A consistent multivariate test of association based on ranks of distances. *Biometrika 100*(2), 503–510.

Lan, Z., Y. Sun, L. Zhao, Y. Xiao, C. Kuai, and S.-W. Xue (2021). Aberrant effective connectivity of the ventral putamen in boys with attention-deficit/hyperactivity disorder. *Psychiatry Investigation 18*(8), 763.

Lei, D., J. Ma, X. Du, G. Shen, X. Jin, and Q. Gong (2014). Microstructural abnormalities in the combined and inattentive subtypes of attention deficit hyperactivity disorder: a diffusion tensor imaging study. *Scientific Reports 4*(1), 1–7.

Li, G., H. Peng, J. Zhang, and L. Zhu (2012). Robust rank correlation based screening.

Li, R., W. Zhong, and L. Zhu (2012). Feature screening via distance correlation learning. *Journal of the American Statistical Association 107*(499), 1129–1139.

Luo, F.-F., J.-B. Wang, L.-X. Yuan, Z.-W. Zhou, H. Xu, S.-H. Ma, Y.-F. Zang, and M. Zhang (2020). Higher sensitivity and reproducibility of wavelet-based amplitude of resting-state fmri. *Frontiers in neuroscience 14*, 224.

Lyons, R. (2013). Distance covariance in metric spaces. *The Annals of Probability 41*(5), 3284–3305.

Meinshausen, N. and P. Bühlmann (2010). Stability selection. *Journal of the Royal Statistical Society: Series B (Statistical Methodology) 72*(4), 417–473.

Pan, W., X. Wang, W. Xiao, and H. Zhu (2019). A generic sure independence screening procedure. *Journal of the American Statistical Association 114*(526), 928.

Pan, W., X. Wang, H. Zhang, H. Zhu, and J. Zhu (2020). Ball covariance: A generic measure of dependence in banach space. *Journal of the American Statistical Association 115*(529), 307–317.

Rubia, K., A. B. Smith, M. J. Brammer, and E. Taylor (2007).

Temporal lobe dysfunction in medication-naive boys with attention-deficit/hyperactivity disorder during attention allocation and its relation to response variability. *Biological psychiatry 62*(9), 999–1006.

Serfling, R. J. (2009). *Approximation theorems of mathematical statistics*. John Wiley & Sons.

Sidlauskaite, J., K. Caeyenberghs, E. Sonuga-Barke, H. Roeyers, and J. R. Wiersema (2015). Whole-brain structural topology in adult attention-deficit/hyperactivity disorder: Preserved global–disturbed local network organization. *NeuroImage: Clinical 9*, 506–512.

Stoodley, C. J. (2016). The cerebellum and neurodevelopmental disorders. *The Cerebellum 15*(1), 34–37.

Székely, G. J., M. L. Rizzo, and N. K. Bakirov (2007). Measuring and testing dependence by correlation of distances. *The annals of statistics 35*(6), 2769–2794.

Wang, J., Z. Zhang, G.-J. Ji, Q. Xu, Y. Huang, Z. Wang, et al. (2015). Frequency-specific alterations of local synchronization in idiopathic generalized epilepsy. *Medicine 94*(32).

Willcutt, E. G. (2012). The prevalence of dsm-iv attention-deficit/hyperactivity disorder: a meta-analytic review. *Neurotherapeutics 9*(3), 490–499.

Wolf, R. C., M. M. Plichta, F. Sambataro, A. J. Fallgatter, C. Jacob, K.-P. Lesch, M. J. Herrmann, C. Schönfeldt-Lecuona, B. J. Connemann, G. Grön, et al. (2009). Regional brain activation changes and abnormal functional connectivity of the ventrolateral prefrontal cortex during working memory processing in adults with attention-deficit/hyperactivity disorder. *Human brain mapping 30*(7), 2252–2266.

Yu-Feng, Z., H. Yong, Z. Chao-Zhe, C. Qing-Jiu, S. Man-Qiu, L. Meng, T. Li-Xia, J. Tian-Zi, and W. Yu-Feng (2007). Altered baseline brain activity in children with adhd revealed by resting-state functional mri. *Brain and Development 29*(2), 83–91.

Zhang, H., L. Zhang, and Y. Zang (2015). Fluctuation amplitude and local synchronization of brain activity in the ultra-low frequency band: an fmri investigation of continuous feedback of finger force. *Brain Research 1629*, 104–112.

Zhang, H., Y. Zhao, W. Cao, D. Cui, Q. Jiao, W. Lu, H. Li, and J. Qiu (2020). Aberrant functional connectivity in resting state networks of adhd patients revealed by independent component analysis. *BMC neuroscience 21*(1), 1–11.

Zhao, Q., H. Li, X. Yu, F. Huang, Y. Wang, L. Liu, et al. (2017). Abnormal resting-state functional connectivity of insular subregions and disrupted correlation with working memory in adults with attention deficit/hyperactivity disorder. *Frontiers in psychiatry 8*, 200.

Zhao, S. D. and Y. Li (2012). Principled sure independence screening for cox models with ultra-high-dimensional covariates. *Journal of multivariate analysis 105*(1), 397–411.

Appendix A : Proof

Proof of Theorem 2.1

Proof. By a U-statistic theory (Serfling, 2009), we have

$$\mathbb{E}[V_{n;m}^k] = \mathbb{E}[h(z_{i_1}^k, ..., z_{i_m}^k; \tau_k)],$$

$$Var[V_{n;m}^k] \leq \frac{m}{n} Var[h(z_{i_1}^k, ..., z_{i_m}^k; \tau_k)]$$

Therefore, for $k \in \mathcal{A}$,

$$\begin{aligned}
\mathbb{E}[V_{n;m}^k] &= P(m\hat{\rho}_{k;m} > \tau_k) \\
&= P(\rho_k - \hat{\rho}_{k;m} < \rho_k - \frac{1}{m}\tau_k) \\
&\geq P(|\rho_k - \hat{\rho}_{k;m}| < \rho_k - \frac{1}{m}\tau_k) \\
&\geq 1 - \exp(-c_2 m(\rho_k - \frac{1}{m}\tau_k)^2)
\end{aligned}$$

where the last inequality is driven from (A2). As $m = c_0 n^\gamma$, we have

$$\exp(-c_2 m(\rho_k - \frac{1}{m}\tau_k)^2) = \mathcal{O}(\exp(-m\rho_k^2)) = \mathcal{O}(\exp(-n^{\gamma-2\kappa})) \to 0.$$

Then

$$\mathbb{E}[V_{n;m}^k] = 1 - \mathcal{O}(\exp(-n^{\gamma-2\kappa}))$$

and

$$\begin{aligned}
Var[h^k(z_{i_1}^k, ..., z_{i_m}^k; \tau_k)] &= \mathbb{E}[h^k](1 - \mathbb{E}[h^k]) \\
&= \mathcal{O}(\exp(-n^{\gamma-2\kappa}))
\end{aligned}$$

We now establish an upper bound of $P(k \notin \hat{\mathcal{A}}_n)$.

(i) Using a Chebyshev inequality,

$$\begin{aligned}
P(k \notin \hat{\mathcal{A}}_n) &= P(V^k_{n;m} < \theta) \\
&= P(\mathbb{E}[V^k_{n;m}] - V^k_{n;m} > \mathbb{E}[V^k_{n;m}] - \theta) \\
&\le \frac{Var(V^k_{n;m})}{2(\mathbb{E}[V^k_{n;m}] - \theta)^2} = \mathcal{O}(n^{\gamma-1}\exp(-n^{\gamma-2\kappa}))
\end{aligned}$$

where the last inequality is derived from

$$Var(V^k_{n;m}) \le \frac{m}{n} Var[h(z^k_{i_1}, ..., z^k_{i_m}; \tau_k)] = \mathcal{O}(n^{\gamma-1}\exp(-n^{\gamma-2\kappa})).$$

(ii) Using a Bernstein bound for U-statistic,

$$P(|V^k_{n;m} - \mathbb{E}[V^k_{n;m}]| > \epsilon) < \exp(-\frac{n/m \cdot \epsilon^2}{2(\sigma^2_{h^k} + \epsilon/3)})$$

where $\sigma^2_{h^k} = \mathbb{E}[h^k](1 - \mathbb{E}[h^k]) = \mathcal{O}(\exp(-n^{\gamma-2\kappa}))$. Then

$$\begin{aligned}
P(k \notin \hat{\mathcal{A}}_n) &= P(V^k_{n;m} < \theta) \\
&= P(\mathbb{E}[V^k_{n;m}] - V^k_{n;m} > \mathbb{E}[V^k_{n;m}] - \theta) \\
&\le \exp(-\frac{n/m \cdot (\mathbb{E}[V^k_{n;m}] - \theta)^2}{2(\sigma^2_{h^k} + (\mathbb{E}[V^k_{n;m}] - \theta)/3}) = \mathcal{O}(\exp(-n^{1-\gamma})).
\end{aligned}$$

Combining (i) and (ii), we have $P(k \notin \hat{\mathcal{A}}_n) = \mathcal{O}(\exp(-n^{(\gamma-2\kappa)\vee(1-\gamma)}))$. Since

$$P(\mathcal{A} \subset \hat{\mathcal{A}}_n) > 1 - p^* P(k \notin \hat{\mathcal{A}}_n),$$

and $log(p^*) = o(n^{(\gamma-2\kappa)\vee(1-\gamma)})$, $P(\mathcal{A} \subset \hat{\mathcal{A}}_n) \to 1$ is derived.

The variance $\sigma^2_{h^k}$ converges to zero, and therefore the Chebyshev inequality provides a better bound than the Bernstein method for some values of γ.

$\square$

Proof of Theorem 2.2

Proof. If $k \in \mathcal{I}$, we use a Bernstein bound for U-statistic;

$$P(|V^k_{n;m} - \mathbb{E}[V^k_{n;m}]| > \epsilon) < \exp(-\frac{n\epsilon^2/m}{2(\sigma^2_{h^k} + \epsilon/3)})$$

where $\sigma^2_{h^k} = \mathbb{E}[h^k](1 - \mathbb{E}[h^k])$. Then

$$\begin{aligned} P(k \in \hat{\mathcal{A}}_n) = P(V^k_{n;m} > \theta) \\ &= P(V^k_{n;m} - \mathbb{E}[V^k_{n;m}] > \theta - \mathbb{E}[V^k_{n;m}]) \\ &\leq \frac{1}{2}\exp(-\frac{n/m \cdot (\theta - \mathbb{E}[V^k_{n;m}]))^2}{2(\sigma^2_{h^k} + (\theta - \mathbb{E}[V^k_{n;m}])/3)}) = \mathcal{O}(\exp(-n^{1-\gamma})) \end{aligned}$$

In a last step, we used $\mathbb{E}[V^k_{n;m}] = \mathbb{E}[h(z^k_{i_1}, ..., z^k_{i_m})] \to \alpha$ and $\sigma^2_{h^k} \to \alpha(1-\alpha)$ so that the order only depends on a term n/m. Since

$$P(\hat{\mathcal{A}}_n \subset \mathcal{A}) = 1 - (p - p^*)P(k \in \hat{\mathcal{A}}_n),$$

given $\gamma < 1$ and $log(p - p^*) = o(n^{1-\gamma})$, we have $P(\hat{\mathcal{A}}_n \subset \mathcal{A}) \to 1$. $\square$

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